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Hiring Committee
School of Engineering / Informatics
University of Edinburgh
Edinburgh, United Kingdom

To the Hiring Committee,

I am writing to formally apply for the fully funded PhD project "Agile SPAD Sensor Image Processing." With my robust background as an AI & Data Science Engineer, specializing in deep learning, high-frequency pipeline optimization, and advanced computer vision, I am highly motivated to contribute to the development of novel Image Signal Processing (ISP) pipelines for SPAD sensors.

My engineering background has heavily focused on the intersection of hardware efficiency and software intelligence. In my recent "Cardiovascular Medical Digital Twin" project, I engineered **Medical_MambaTS**, an architecture specifically designed for sub-millisecond, linear-time inference directly at the edge. To support this, I built a zero-copy PyArrow data pipeline that prevents GPU starvation and minimizes memory overhead. This direct experience in optimizing low-latency data streams translates perfectly to your objective of designing a bespoke ISP optimized for high-speed, single-photon input data.

Furthermore, deploying these SPAD sensors into real computer vision applications such as robotic navigation and tracking perfectly aligns with my computer vision portfolio. I have extensive experience in spatio-temporal tracking, multimodal reasoning, and deep feature extraction using PyTorch. The challenge of designing dynamic vision systems—such as refining optimal ISP partitioning and in-pixel computation—is an area where my dual expertise in physics-informed AI (enforcing dynamic constraints via PI-DeepONet) and systems-level pipeline engineering will allow me to rapidly produce tangible demonstration systems.

I am deeply drawn to the University of Edinburgh's pioneering research in high-performance computing and quantum-influenced sensor applications. Enclosed are my CV, a description of my relevant technical projects, and my academic transcripts. I would welcome the opportunity to discuss how my expertise can accelerate the deliverables of this SPAD sensor image processing project.

Sincerely,

Ismail Aissaoui